SMART INDIA HACKATHON 202

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TITLE PAGE

- **Problem Statement ID –** SIH26146
- **Problem Statement Title-** AI-Powered Monitoring & Analysis of Bitcoin Transaction Traffic
- **Theme-** Transportation & Logistics
- **PS Category- Software**
- **Team ID-**
- **Team Name (Registered on portal)**

Your Team Name

IDEA TITLE

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❖ Proposed Solution (Describe your Idea/Solution/Prototype)

- Links blockchain-level data (wallets, transactions) and network-level data (P2P/IP signals) into one graph; AI/ML flags unusual patterns; output is a short, prioritized, plain-language list of leads — not raw transaction dumps.
- Solves the core bottleneck: investigators can’t manually connect fragmented on-chain and network signals at scale.
- That gap is widening fast: FIU-IND logged a 773% jump in crypto-related suspicious transaction reports (1,343 in FY24 → 11,720 in 8 months of FY26).
- A sovereign, cost-effective, fully auditable alternative to foreign commercial forensics tools — in-house and under Indian agency control end-to-end. vs. Chainalysis/Elliptic/TRM Labs: no publicly disclosed network-layer fusion or plain-language output, and foreign-hosted.

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TECHNICAL APPROACH

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- Python (NetworkX, PyTorch Geometric, scikit-learn) · FastAPI · Neo4j/PostgreSQL · React+Cytoscape.js · BigQuery public Bitcoin dataset. Model: GraphSAGE/GCN pretrained on Elliptic-labeled nodes, transferred to the fused graph; evaluated via precision/recall against Elliptic ground truth.

Data Sources
Ledger + P2P node
→
Ingestion & Graph
Unify into 1 graph
→
Feature Eng.
Patterns, timing
→
GraphSAGE/GCN
Elliptic-pretrained
→
Explainable Output
Ranked, reasoned

Hackathon demo: simulated/replayed P2P signal. Caveat: BIP 324 encrypts most Bitcoin P2P traffic by default (Core v27+) — correlation now requires probe nodes peering directly with a target (not passive listening), needing ~25%+ of its connection slots for usable signal (NTSSL: 50% precision/56.8% recall; NTSSL+: ~60%/72.7%, Zhang et al., 2026). Live deployment is post-Round-1, requiring lawful-intercept-equivalent authority via NTRO.

Wallet 1A2b...E4f9 — Risk: 87/100
Reason: 3 hops from a flagged address;
unusual off-hours transaction burst

Explanation method: rule-based post-hoc overlay translating top contributing graph features (hop-distance to a flagged node, timing-anomaly score) into templated text — not the GNN's raw output.

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FEASIBILITY AND VIABILITY

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MVP is feasible within a 36-hour build: BigQuery's public Bitcoin dataset avoids running a full node; the demo trains a lightweight GCN baseline (full transfer-learning pipeline is a Round-2 target).

Challenges

- CoinJoin/mixing defeats simple co-spend clustering
- False positives carry real privacy consequences
- Network-layer correlation is probabilistic, not deterministic
- Full-chain analysis is resource-heavy
- Elliptic dataset (2019) reflects older transaction patterns

Mitigations

- Hybrid-motif mixer-detection (Wu et al., 2021) flags mixed txns as low-confidence
- Every flag carries a confidence score for human review — never auto-action
- Every flag logs its derivation chain for auditability
- Network monitoring assumes lawful-intercept-equivalent authority under judicial/statutory oversight
- Periodic retraining + synthetic augmentation planned to offset dataset age

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IMPACT AND BENEFITS

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- Directly supports FIU-IND, the Enforcement Directorate, and state cyber cells — whose crypto-related suspicious transaction caseload rose 773% (1,343 in FY24 → 11,720 in 8 months of FY26, Hindustan Times, citing FIU-IND analysis).
- Reclaims investigator time by triaging leads instead of raw data — illustrative estimate: ~1,170 investigator-hours/year (11,720 cases × 10% slice × ~2 hrs/case baseline × 50% reduction). Bitcoin is the pilot ledger; architecture generalizes to the stablecoin/VASP majority of flagged cases (Tether ~76%, Bitcoin ~6% of FIU-IND's crypto STRs). Supports India's AML posture under the 2023 PMLA amendment and FATF Rec. 15.

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RESEARCH AND REFERENCES

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- Weber et al. (2019), arXiv:1908.02591 — Elliptic dataset & GCN baseline
- Meiklejohn et al. (2013), ACM IMC — co-spend clustering heuristics
- Kaminsky (2011), Black Hat — first-spy P2P deanonymization concept; Mehta, Ruffing, Schnelli & Wuille, BIP 324 — v2 P2P encrypted transport spec, default since Bitcoin Core v27.1
- Biryukov, Khovratovich & Pustogarov (2014), ACM CCS — Bitcoin P2P client deanonymization
- Wu et al. (2021), IEEE TSMC — mixing-service detection via hybrid motifs
- Ghimire, Murad & Rahimi (2026), arXiv:2603.16080 — Euclidean vs. hyperbolic GNNs on Bitcoin graphs
- GraphSense (open-source blockchain analytics); FIU-IND suspicious transaction report data, FY23-24 to FY25-26; Zhang, Han, Tian, Shi, Lan & Wang (2026), arXiv:2603.17261 — recent network-traffic deanonymization research (NTSSL)